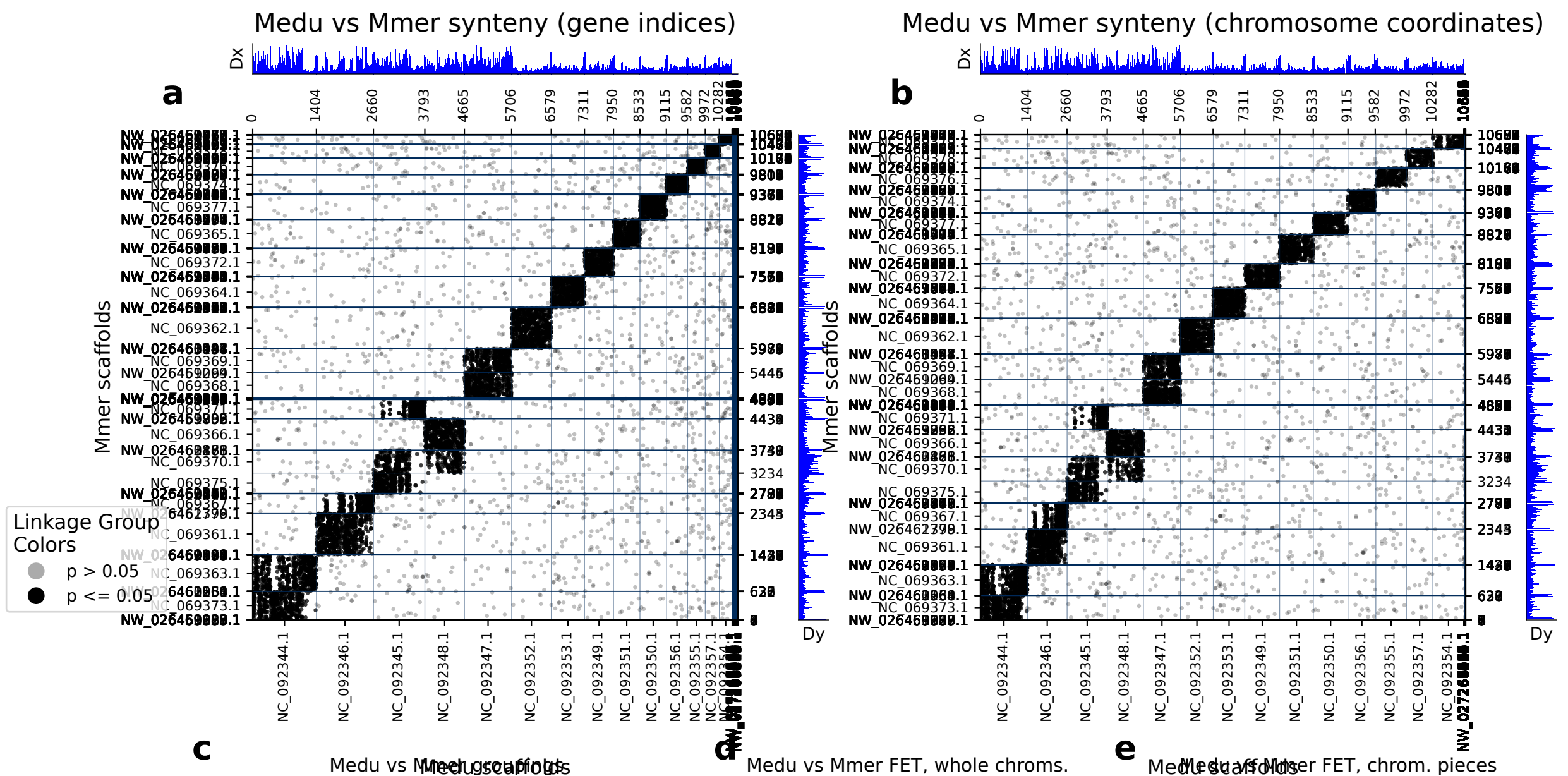


diamond: 10692 orthologs on 129 Medu scaffolds and 259 Mmer scaffolds.



	Medu scaf	Mmer scaf	p (FET)	Count
0	NC_092352.1	NC_069362.1	0.00e+00	748
1	NC_092346.1	NC_069361.1	0.00e+00	747
2	NC_092344.1	NC_069363.1	0.00e+00	667
3	NC_092348.1	NC_069366.1	0.00e+00	562
4	NC_092353.1	NC_069364.1	0.00e+00	554
5	NC_092344.1	NC_069373.1	0.00e+00	517
6	NC_092349.1	NC_069372.1	0.00e+00	486
7	NC_092347.1	NC_069368.1	0.00e+00	472
8	NC_092351.1	NC_069365.1	0.00e+00	463
9	NC_092350.1	NC_069377.1	0.00e+00	436
10	NC_092347.1	NC_069369.1	9.47e-315	400
11	NC_092345.1	NC_069375.1	3.67e-265	349
12	NC_092356.1	NC_069374.1	0.00e+00	322
13	NC_092346.1	NC_069367.1	8.55e-211	320
14	NC_092345.1	NC_069371.1	5.83e-224	313
15	NC_092355.1	NC_069376.1	0.00e+00	255
16	NC_092345.1	NC_069370.1	1.33e-95	234
17	NC_092357.1	NC_069378.1	5.93e-248	192
18	NC_092348.1	NC_069370.1	1.31e-57	167
19	NC_092354.1	NC_069379.1	7.80e-161	128

	Medu scaf	Mmer scaf	p (FET)	Count
0	NC_092352.1 : 0-end	NC_069362.1 : 0-end	0.00e+00	748
1	NC_092346.1 : 0-end	NC_069361.1 : 0-end	0.00e+00	747
2	NC_092344.1 : 0-end	NC_069363.1 : 0-end	0.00e+00	667
3	NC_092348.1 : 0-end	NC_069366.1 : 0-end	0.00e+00	562
4	NC_092353.1 : 0-end	NC_069364.1 : 0-end	0.00e+00	554
5	NC_092344.1 : 0-end	NC_069373.1 : 0-end	0.00e+00	517
6	NC_092349.1 : 0-end	NC_069372.1 : 0-end	0.00e+00	486
7	NC_092347.1 : 0-end	NC_069368.1 : 0-end	0.00e+00	472
8	NC_092351.1 : 0-end	NC_069365.1 : 0-end	0.00e+00	463
9	NC_092350.1 : 0-end	NC_069377.1 : 0-end	0.00e+00	436
10	NC_092347.1 : 0-end	NC_069369.1 : 0-end	9.47e-315	400
11	NC_092345.1 : 0-end	NC_069375.1 : 0-end	3.67e-265	349
12	NC_092356.1 : 0-end	NC_069374.1 : 0-end	0.00e+00	322
13	NC_092346.1 : 0-end	NC_069367.1 : 0-end	8.55e-211	320
14	NC_092345.1 : 0-end	NC_069371.1 : 0-end	5.83e-224	313
15	NC_092355.1 : 0-end	NC_069376.1 : 0-end	0.00e+00	255
16	NC_092345.1 : 0-end	NC_069370.1 : 0-end	1.33e-95	234
17	NC_092357.1 : 0-end	NC_069378.1 : 0-end	5.93e-248	192
18	NC_092348.1 : 0-end	NC_069370.1 : 0-end	1.31e-57	167
19	NC_092354.1 : 0-end	NC_069379.1 : 0-end	7.80e-161	128

(a) depicts the chromosomes/scaffolds of Medu (x-axis) plotted against the chromosomes/scaffolds of Mmer (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10692 orthologs found between 129 Medu scaffolds and 259 Mmer scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. *Nature Ecology & Evolution* 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.